

is transferred to the target via email, suspicious links, USB media, etc.

4. **Exploitation:** Once the executable is delivered to the victim, the payload triggers the Trojan and exploits the operating system's vulnerability or targets a specific app.
5. **Installation:** Allows the Trojan and attacker to remain in close proximity.
6. **Command and control server:** A command and control server is used to control the Trojan installed in the target via remote access.
7. **Actions and goals:** After six stages of progress, the goal is accomplished by compromising the host.

The intrusion kill chain often takes many days for a successful attack to take place. It is very hard to identify these chain processes due to their 'stepping stone' behaviour (compromising multiple hosts to reach the target). It's very difficult to identify the first three stages of the attack cycle.

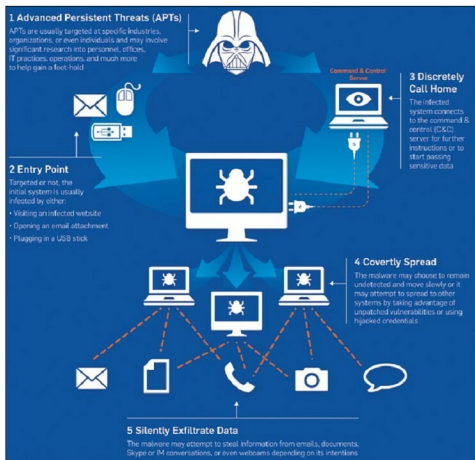


Figure 1: APT anatomy

APT vs traditional malware

Traditional viruses/ malware	APT
Targets random networks/ hosts	Targets specific networks/ hosts
Probability of getting detected by antivirus is high as signatures get detected	A combination of malware is used. Signatures go undetected because of this
The effects become visible over a period of time, as large networks/hosts get infected	The idea is to lie low over a significant period of time
A good firewall or intrusion detection system can prevent entry by signature checking	It is carried mostly through content, which uses well known ports (80, 443, etc) and known protocols like http, https, etc

Zero day exploits

A zero day exploit is a computer application vulnerability which is unpatched and previously not known to security vendors. These security holes are exploited by the attacker before the vendor becomes aware of the exploits. Zero day attacks include data infiltration and exfiltration malware, malicious code, etc. The best example of a zero day exploit is *Stuxnet*, which was named 'the first digital weapon'.

Stuxnet

Stuxnet, the first digital weapon, ranks No. 1 among zero day

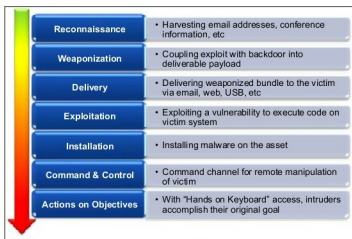


Figure 2: Various stages of the intrusion kill chain

vulnerabilities when it comes to causing collateral damage to computers in the real world. Stuxnet was first discovered in June 2010, and reportedly destroyed one-fifth of the centrifugal pumps of the Iranian nuclear facility. Stuxnet is specific to PLC boards and targets the rotator which spins them up and down. It has unique characteristics which differ from traditional payloads of various malware and Trojans. According to the Kaspersky report, once the Trojan hits the system, it automatically compromises the host PLC boards and sparks chaos.

Components of Stuxnet

Various aspects of Stuxnet differentiate it from other